Orbital Dragons to Hunt Cryptids: Utilizing Non-Euclidean Geometry to Solve Busy Beaver

Enan Srivastava

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The busy beaver value $BB(n)$ is the longest finite runtime of any n-state Turing machine started on a blank tape. $BB(5)$ was settled in 2024 with a computer-checked proof. $BB(6)$ is blocked by a handful of machines whose tapes secretly run Collatz-style arithmetic; nobody can prove whether they stop. This paper studies two of these machines and surveys the rest. The approach is empirical first: the machine's behavior is logged as a sequence of events, the event data embedded in logarithmic, simplex, and hyperbolic coordinates, and the dynamics found to be piecewise affine in a five-integer state. The description is then proven correct in the Lean proof assistant, down to the machine's individual steps and with a minimal axiom footprint. The result is a machine-checked theorem: if a simple five-integer recurrence never enters an explicit finite set of bad configurations, the machine never halts. The converse pressure is also proven: entering a bad configuration forces the machine to halt. The open problem is thereby moved, without loss, from a Turing machine to a short arithmetic recurrence. The renewal map of the recurrence is then derived in closed form and shown to have a width-five fatal window around the diagonal, and the two mechanical proof routes with computational evidence: no congruence invariant up to modulus 384 excludes the window, and the branch dynamics has exponential factor complexity, so it is not conjugate to a rotation. Measuring the recurrence over its whole seed space shows the set of dying seeds has natural density zero at the Borel–Cantelli rate ($\approx 12/X$), so the studied machines' seeds are typical survivors. Surveying the entire holdout set with a two-ended harness (forward orbit against backward halt cone, regimes fitted in between) classifies all 1,094 open six-state machines: zero halts, zero alignments; four fixed-dimension machines, exactly the certified ones; an eleven-machine *array class* whose state spaces grow in dimension, for whose leading member an exact word-rewrite calculus is constructed (a two-phase odometer with a block-trisection boundary rule) validated against 2×10^8 raw steps; and 307 single-law machines among 346 with regime structure. The program is framed on a verification spectrum whose proven endpoint is $\mathsf{MIP}^* = \mathsf{RE}$: kernel checking at the bottom, certified predictive emulation (this paper) in the middle, and entangled-prover verification of halting-type claims as the goal. Every empirical claim is validated against 10^9 raw machine steps, and the recurrence is tested for 10^8 iterations.

The central certified statement, one per machine:

Theorem 1. *Checked by the Lean kernel at axioms* {propext, Quot.sound}*:* $\mathsf{SafeMetaOrbit} \to \neg\mathsf{RawHalts}(\mathsf{blankStart})$*, and configurations outside the guarded region halt within about fifty steps, so* SafeMetaOrbit *is the halting problem restated as arithmetic.*

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